

Unit 3: Instruction Cycle

Instruction cycle, machine cycle and T states

Machine cycle: Machine cycle is the time required to transfer data to or from memory or I/O devices. It is defined as the time required to complete one operation of accessing memory, i/p, o/p or acknowledging and external request. This cycle may consist of 3 to 6 T states.

T-states: Each machine cycle consists of many clock periods/cycles, called T-states. Each and every operation inside the microprocessor is under the control of the clock cycle. The clock signal determines the time taken by the microprocessor to execute any instruction. The clock cycle shown in Fig. below has two edges (leading and trailing or lagging). T-State is defined as the time interval between 2-trailing or leading edges of the clock. It is a sub-division of the operation performed in one clock period. These sub-divisions are internal states synchronized with system clock and each T state is precisely equal to one clock period.

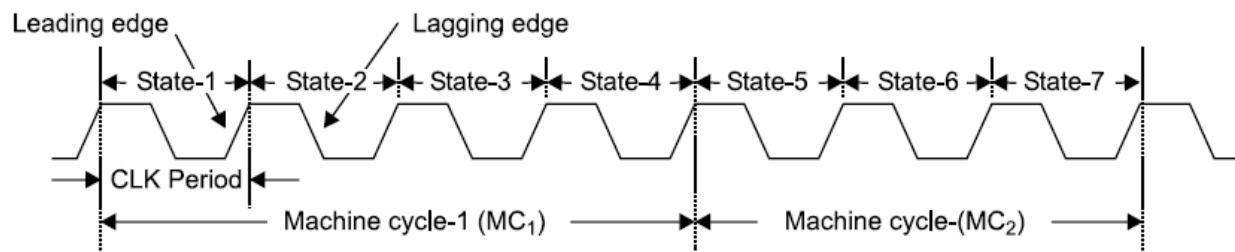


Fig. (a) Machine cycle showing clock periods

Instruction cycle: The necessary steps that the CPU carries out to fetch an instruction and necessary data from the memory and to execute it constitute an instruction cycle. It is defined as the time required to complete the execution of an instruction. An instruction cycle consists of a fetch cycle and an execute cycle. In the fetch cycle, the CPU fetches the opcode from the memory. The necessary steps which are carried out to fetch an opcode from memory constitute a fetch cycle. The necessary steps which are carried out to get data if any from the memory and to perform the specific operation specified in an instruction constitute an execute cycle. The total time required to execute an instruction is given by $IC = Fc + Ec$. The 8085 consists of 1-6 machine cycles or operations.

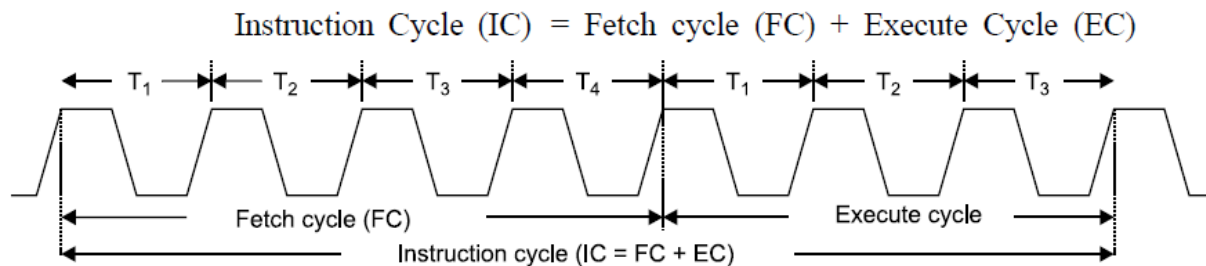
Fetch cycle: The first byte of an instruction is its opcode. The program counter keeps the memory address of the next instruction to be executed. In the beginning of the fetch cycle, the content of the program counter, which is the address of the memory location where the opcode is available, is sent to the memory. The memory places the opcode on the data bus so as to transfer it to the CPU. The entire process takes 3 clock cycles.

Execute cycle/Operation:- The opcode fetched from the memory goes to the IR. From the IR, it goes to the decoder which decodes the instruction. After the instruction is decoded, execution begins. If the operand is in a general-purpose register, execution is performed immediately, i.e. in one clock cycle. If an instruction contains data or an operand address, then the CPU has to perform some read operations to get the desired data. In some instructions, a write operation is performed. In a write cycle, data are sent from the CPU to the memory of an I/O device. In some cases, the execute cycle may involve one or more read or write cycles or both.

Machine cycle of 8085 microprocessor: Opcode fetch, Memory read, memory write, I/O read, I/O write, Interrupt

The function of the microprocessor is divided into fetch and executes cycle of any instruction of a program. The program is nothing but number of instructions stored in the memory in sequence. In the normal process of operation, the microprocessor fetches (receives or reads) and executes one instruction at a time in the sequence until it executes the halt (HLT) instruction. Thus, an instruction cycle is defined as the time required fetching and executing an instruction. For executing any program, basically 2-steps are followed sequentially with the help of clocks **Fetch and Execute**.

The time taken by the μP in performing the fetch and execute operations are called fetch and execute cycle. Thus, sum of the fetch and execute cycle is called the instruction cycle as indicated in Fig



The 1st machine cycle (M1) of every instruction cycle is the op-code fetch cycle. In the op-code fetch cycle, the processor comes to know the nature of the instruction to be executed. The processor during (M1 cycle) puts the program counter contents on the address bus and reads the opcode of the instruction through read process. The T1, T2, and T3 clock cycles are used for the basic memory read operation and the T4 clock and beyond are used for its interpretation of the opcode.

Based on these interpretations, the μP comes to know the type of additional information/data needed for the execution of the instruction and accordingly proceeds further for 1 or 2-machine cycle of memory read and writes. The Op. code fetch cycle is of fixed duration (normally 4-states), whereas the instruction cycle is of variable duration depending on the length of the instruction. As an example, STA instruction, requires opcode fetch cycle, lower-order address fetch cycle and higher order fetch cycle and then the execute cycle.

One machine cycle is required each time the μP access I/O port or memory. A fetch opcode cycle is always 1-machine cycle, whereas, execute cycle may be of one or more machine cycle depending upon the length of the instruction. The instruction is decoded and translated into specific activities during the execution phase. Duration of machine cycle varies according to microprocessor architecture.

Opcode Fetch operation

A microprocessor either reads or writes to the memory or I/O devices. The time taken to read or write for any instruction must be known in terms of the μP clock. The 1st step in communicating between the microprocessor and memory is reading from the memory. This reading process is called opcode fetch. The 1st machine cycle of any instruction is always the fetch cycle that provides identification of the instruction to be executed. The process of opcode fetch operation requires minimum 4- clock cycles T1, T2, T3, and T4 and is the 1st machine cycle (M1) of every instruction. In order to differentiate between the data byte pertaining to an opcode or an address, the machine cycle takes help of the status signal IO/M, S1, and S0. The IO/M = 0 indicates memory operation and S1 = S0 = 1 indicates Opcode fetch operation. The opcode fetch machine cycle M1 consists of 4-states (T1, T2, T3, and T4). The 1st 3- states are used for fetching (transferring) the byte from the memory and the 4th-state is used to decode it.

Figs. below depict these first-steps for implementation of the instruction **ADI 05H**. Let us assume that the accumulator contains the result of previous operation i.e., **03H** and instruction is held at memory locations **2030H** and **2031H**.

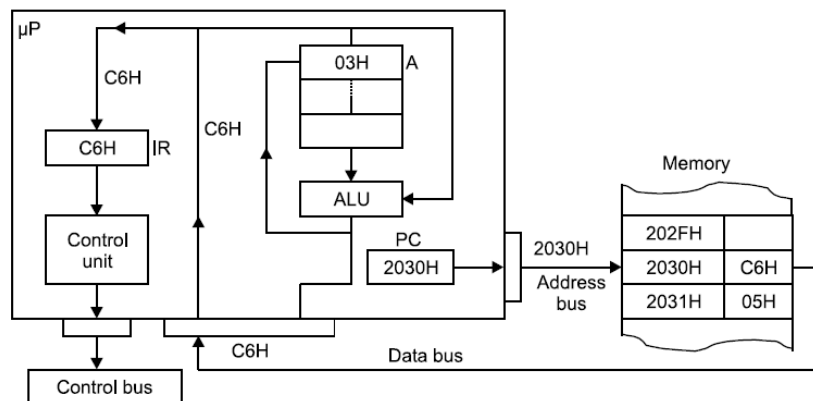


Fig. Instruction fetch : reads 1st byte (Opcode) in instruction register (IR)

The fetch part of the instruction is the same for every instruction. The control unit puts the contents of the program counter (PC) **2030H** on the address bus. The 1st byte (op-code **C6H** in this example) is passed to the instruction register.

Execute Operation:

In the execute cycle of the instruction, the control unit examines the op-code and as per interpretation further memory read or write operations are performed depending upon whether additional information/ data are required or not.

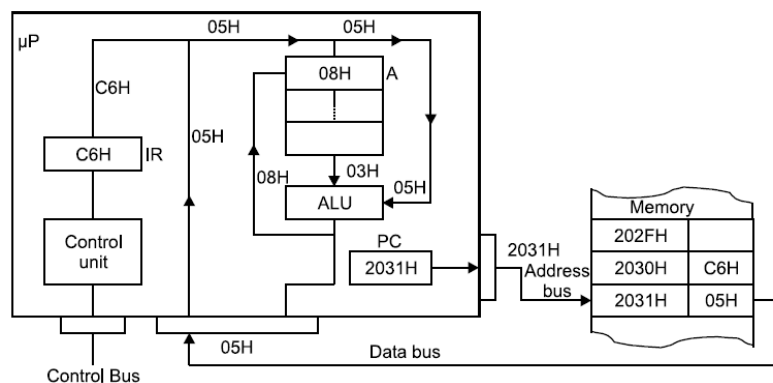


Fig. Instruction execute : reads 2nd byte from memory and adds to accumulator

In this case, the data **05H** from the memory is transferred through the data bus to the ALU. At the same time the control unit sends the contents of the accumulator (**03H**) to the ALU and performs the addition operation. The result of the addition operation **08H** is passed to the accumulator overriding the previous contents **03H**. On the completion of one instruction, the program counter is automatically incremented to point to the next memory location to execute the subsequent instruction.

Timing diagram of MOV, MVI, IN, OUT, LDA, STA

Timing diagram: The necessary steps which are carried in a machine cycle can be represented graphically. Such graphical representation is called timing diagram. Timing diagram is the display of initiation of read/write and transfer of data operations under the control of 3-status signals IO / M , S₁, and S₀.

The 8085 microprocessor is designed to execute 74 different instruction types. Each instruction has two parts: operation code, known as op-code, and operand. To execute an instruction, the 8085 needs to perform various operations such as Memory Read/Write and I/O Read/ Write. The total operations of a microprocessor can be classified into the following operations.

- Op-code Fetch
- Memory Read and Write
- I/O Read and Write
- Request Acknowledge

Timing Diagram for Opcode fetch

The following steps occur during Op-code Fetch Cycle

1. A low IO/M means microprocessor wants to communicate with memory. The Program counter (PC) places the 16 bit memory address on the address bus. At T₁ high order address is placed at A₈-A₁₅ and lower order address is placed at AD₀-AD₇ and the ALE signal goes high, both s₀ and s₁ goes high; which identifies the op-code fetch cycle.
2. The control unit sends the control signal RD to enable the memory chip and remains active till two clock periods.
3. Now the op-code from memory location is placed on the multiplexed data bus. In this case 3E is placed on AD₇-AD₀. The transition of RD indicates the end of this step.
4. The op-code byte is now placed on instruction decoder and the execute cycle is carried out.

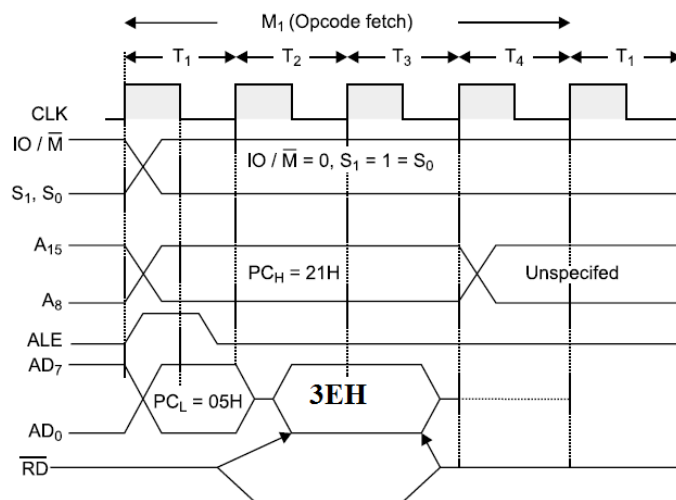


Fig. Opcode fetch

Example : Timing Diagram for MOV B,C.

Assume memory location as below.

Mnemonics	Machine code	Memory Locations
MVI B, 05H	06H	1000H
	05H	1001H

T1: The 1st clock of 1st machine cycle (M1) makes ALE high indicating address latch enabled which loads low-order address 00H on AD7 $\Leftrightarrow$ AD0 and high-order address 10H simultaneously on A15 $\Leftrightarrow$ A8. The address 00H is latched in T1.

T2: During T2 clock, the microprocessor issues RD control signal to enable the memory and memory places 41H from 1000H location on the data bus.

T3 : During T3, the 41H is placed in the instruction register and RD = 1 (high) disables signal. It means the memory is disabled in T3 clock cycle. The opcode cycle is completed by end of T3 clock cycle.

T4 : The opcode is decoded in T4 clock and the action as per 41H is taken accordingly. In other word, the content of C-register is copied in B-register.

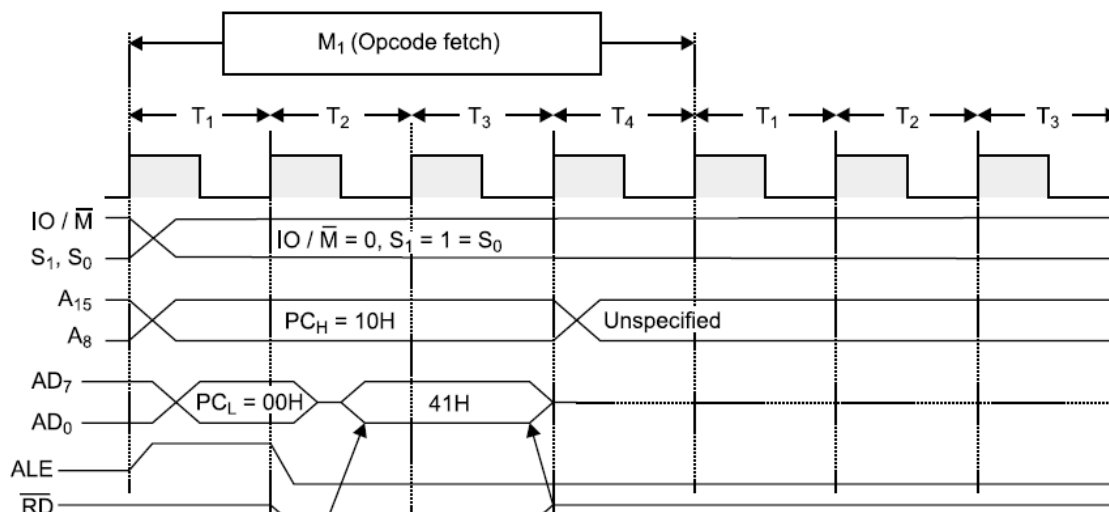


Fig. Opcode fetch (MOV B,C)

Timing Diagram for Memory Read /Write

Memory Read cycle: The Op-code fetch cycle is a memory read cycle. Thus, other memory read cycles are similar to the op-code fetch cycle. Let us take the same example as above.

EXAMPLE Timing diagram of MVI B, 05H

The instruction cycle of **MVI B, 05H** is shown below: The total cycle consist of **7 T states** and **2 machine cycles**: **op-code fetch** and **memory read**. At the end of op-code fetch the PC is incremented thus the address is now **1001h** and instruction decoder has **06h**. Now the operand is to be read from the memory to register B. The second machine cycle is the memory read and consists of **3 T states**. The signal content of this cycle are similar to the first three T states of op-code fetch except the status signal, In this case **S0=0** and **S1=1**. The byte read in **T3** of memory read cycle will be copied into the accumulator in the same cycle.

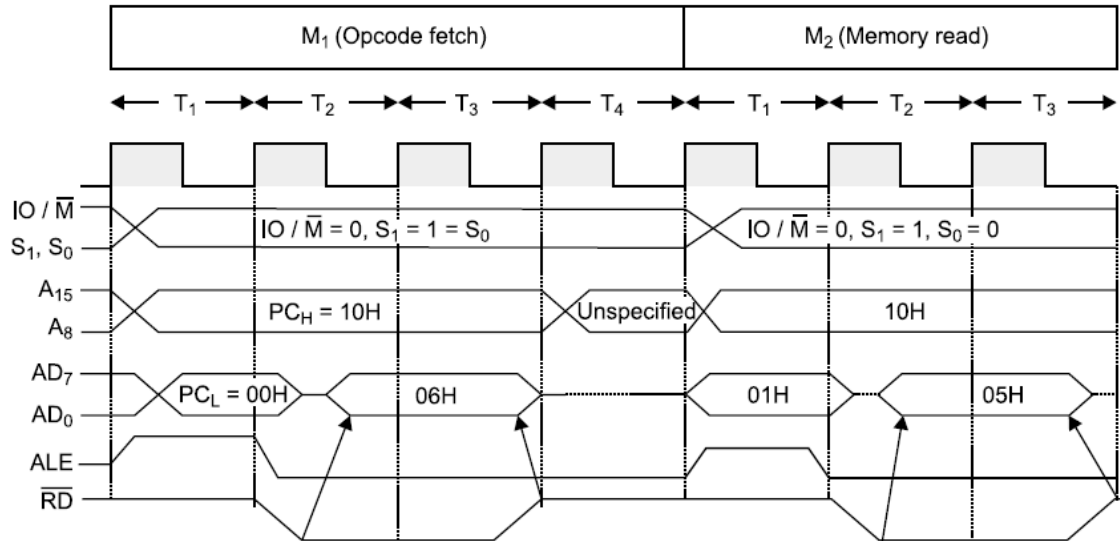


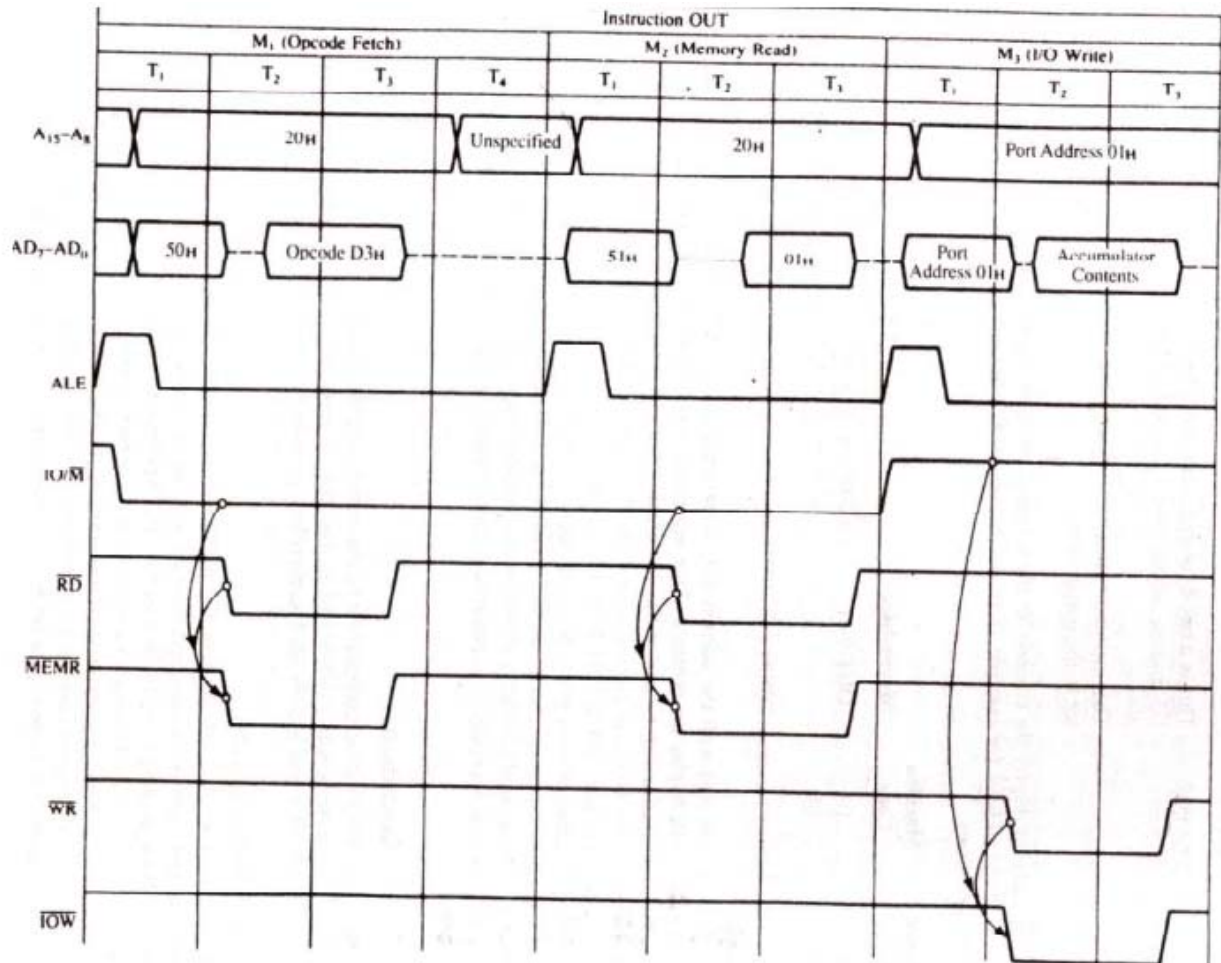
Fig. Timing diagram for MVI B,05H

The **MVI B, 05H** instruction requires 2-machine cycles (M1 and M2). M1 requires 4-states and M2 requires 3-states, total of 7-states as shown in Fig. above. Status signals IO/M, S₁ and S₀ specifies the 1st machine cycle as the op-code fetch. In T₁-state, the high order address {10H} is placed on the bus A₁₅ ⇌ A₈ and low-order address {00H} on the bus AD₇ ⇌ AD₀ and ALE = 1. In T₂ -state, the RD line goes low and the data 06 H from memory location 1000H are placed on the data bus. The fetch cycle becomes complete in T₃-state. The instruction is decoded in the T₄-state. During T₄-state, the contents of the bus are unknown. With the change in the status signal, IO/M = 0, S₁ = 1 and S₀ = 0, the 2nd machine cycle is identified as the memory read. The address is 1001H and the data byte [05H] is fetched via the data bus. Both M1 and M2 perform memory read operation, but the M1 is called op-code fetch i.e., the 1st machine cycle of each instruction is identified as the op-code fetch cycle.

Memory write cycle: It is similar to read cycle except that instead of activating RD(Read) signal, WR(Write) signal is activated as well as S₀=1, S₁=0 after address is latched for sending data in memory.

Input / Output Read Write

I/O Write Cycle: Let us consider the instruction OUT 01h stored at memory location 2050h. The op-code of this instruction is D3h and the complete cycle requires 10 T states. The write cycle executes as shown below. The instruction is executed under three machine cycles: op-code fetch, memory read and I/O write. The op-code fetch and memory read cycles exactly match the previous ones except for the contents of the buses. At the end of memory read cycle the PC points at 2051h. Now instruction decoder contains D3h and on the next machine cycle the port address 01h is placed on higher and lower port address. Unlike previous cases now IO/M signal goes high to indicate a I/O operation. At T₂ the accumulator contents are placed on the data bus, followed by the control signal WR. The signal IOW is obtained by anding the IO/M and WR signals to enable the output device



I/O Read Cycle:

The I/O read cycle is exactly same as I/O write cycle. The only difference is on the third machine cycle where the signal $\overline{IOR}$ is generated and the content of the port is read to the accumulator instead of otherwise. The $\overline{IOR}$ signal is generated by anding the $IO/\overline{M}$ and $\overline{RD}$ signals to enable the input device.

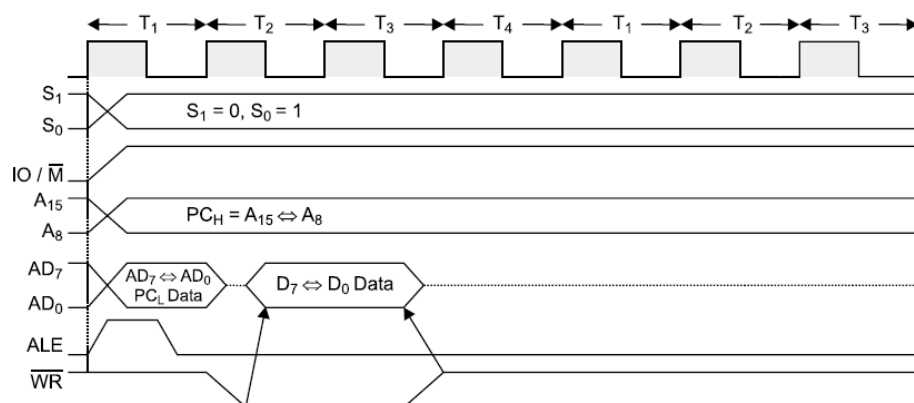


Fig. I/O write timing diagram

Fetch and execute overlap

During fetch the instruction is read from its memory location into the processor. After the instruction is fetched it ought to be executed. The time required for execution is the execution cycle. The fetch and execute stages of instruction execution are quite independent. The fetch cycle involves accessing the main memory for instruction. But during execution the memory need not be referenced. So, during this duration the next instruction can be fetched. This is called instruction **prefetch**. This prefetching data before it is asked for in the anticipation that it will be used is called **pipelining**. The process of pipelining enhances the speed of processor cycle and execution cycle.

